Module 10 Lecture - Introduction to Experiments

Research Methods in Psychology

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Table of Contents

1 Week Overview **2**
1.1 Learning Objective(s) . . . 2
1.2 Week Overview . . . 2

2 Experimental Variables **2**
2.1 Overview . . . 2
2.2 Independent & Dependent . . . 3
2.3 Control . . . 4

3 Why are Experiments Causal? **5**
3.1 Overview . . . 5
3.2 Covariance . . . 5
3.3 Temporal Precedence . . . 6
3.4 Internal Validity . . . 6
3.4.1 Variability . . . 7
3.4.2 Selection Effects . . . 7

4 Independent-groups Design **8**
4.1 Overview . . . 8
4.2 Posttest-only . . . 8
4.3 Pretest/Posttest . . . 8
4.4 Comparison . . . 9

5 Within-groups Design **10**
5.1 Overview . . . 10
5.2 Repeated Measures . . . 10
5.3 Concurrent Measures . . . 10
5.4 Causal Criteria in Within-Groups . . . 10
5.4.1 Internal Validity . . . 11
5.5 Advantages . . . 12
5.6 Disadvantages . . . 12
5.7 Pretest-posttest vs Repeated Measures . . . 13

6 Four Validities Critique **14**
6.1 Construct Validity . . . 14
6.2 External Validity . . . 14
6.3 Statistical Validity . . . 15
6.4 Internal Validity . . . 15

1 Week Overview

1.1 Learning Objective(s)

- Apply the three criteria for establishing causation to experiments and explain why experiments can support causal claims.
- Identify an experiment's independent, dependent, and control variables.
- Classify experiments as independent-groups and within-groups designs and explain why researchers might conduct each type of study.
- Evaluate three potential threats to internal validity in an experiment— design confounds, selection effects, and order effects—and explain how experimenters usually avoid them.
- Interrogate an experimental design using the four validities.

1.2 Week Overview

- Experiments are _____ investigations that are supportive of _____ claims.
- They tell us about _____ influence that one variable has upon another
- They also tend to be the most _____ and difficult studies, with lots of required resources

2 Experimental Variables

2.1 Overview

- Experiments will have at least one measured variable and at least one _____ variable
- *Example (from the first test):* Changing pillow type (manipulated variable) results in change in sleep quality (measured variable)

Discuss: What word in the prior example should really cue you to this being a casual claim?

- *Review:* a manipulated variable is one that we can directly influence and _____, to have a hypothesized impact on our measured variables
- *Review:* measured variables will be things like self-reports or _____ measures

2.2 Independent & Dependent

- An **independent variable** is one that we can change "independent" of other factors, i.e., it does not _____ on something else
  - The manipulated variable
  - It levels (i.e., possible values) are called _____
- A **dependent variable** is one that is somehow hypothesized to be connected to or _____ upon the condition or state of the independent variable
  - The measured variable
  - It may also be referred to as an outcome variable, similar to a multiple regression design
- Often, the _____ variable will be categorical/grouped and the dependent variable will be continuous - but this is not a hard and fast rule

? I am looking at whether depression symptoms are improved more by psychotherapy or medication treatments - what construct is being treated as dependent?

- (●) A) Depression symptoms
- ( ) B) Medication
- ( ) C) psychotherapy
- ( ) D) None of these

Explanation:

2.3 Control

- **Control variables** are those that are _____ held at the same level across participants, to prevent it from having an impact on the outcome

Discuss: What do we call some value that does not change values in our data?

- The purpose of keeping these still is to effectively _____ the effect of the independent variable(s), so that it does not confound our causal link.
- *Example:* I am looking at whether more exercise leads to an increase in energy. However, I suspect that caffeine level could confound this relationship, so I instruct participants to not take any caffeine. Caffeine level across participants is consistently 0, therefore it is a control variable and constant.

? What do we call a variable that interferes with internal validity, like in the prior example?

- (●) A) A sample
- ( ) B) A covariate
- ( ) C) A third variable
- ( ) D) A criterion

Explanation:

3 Why are Experiments Causal?

3.1 Overview

- *Review:* Recall our three causal criteria
  - _____
  - Temporal precedence
  - Internal validity
- Finally, with experiments we have a _____ by which to definitively establish all of these

3.2 Covariance

- If our separate _____ see differences in the outcome variable (i.e., mean differences on a continuous outcome), we would say there is covariance between the independent and dependent variables
- We have to distinguish what conditions we may assign or manipulate participants to:
  - **Comparison groups:** We have at least _____ conditions or levels in our independent variable, so that we can compare them

? What source of information did we explicitly say lacks comparison groups, early in the semester?

- (●) A) Research
- ( ) B) Personal experience
- ( ) C) Authority
- ( ) D) Intuition

Explanation:

- We may use specific labels for types of comparison groups:
  - **Control group:** One that receives no or _____ intervention
  - **Treatment group:** One that receives the intervention or change of interest
  - **Placebo group:** A type of control, that is exposed to an inert or _____ treatment

- Covariance is _____ as important as the other causal criteria
  $\longrightarrow$ no difference between conditions = no causation
- *Example:* I try two different teaching styles between classes to deliver a particular lesson. I take a test at the end of the class on the content. If there is a difference in test scores between the teaching styles, there is covariance.

3.3 Temporal Precedence

- By _____ the independent variable prior to measuring the dependent variable, we establish that it came prior and that we have temporal precedence.
- We will discuss how exactly to implement time in our discussions on the design types shortly

3.4 Internal Validity

- This is the most _____ and core feature that sets experiments apart from other designs, and satisfies the last criteria for causality
- Put simply, we must rule out other reasonable and theoretically-salient _____ for the effects that we see in the outcome variable
- **Confounding variables/confounds:** Those that offer alternative _____ for the effects seen in an outcome.
  - Often we will try to _____ possible confounding variables
  - When a confound results from poor planning in the research design $\longrightarrow$ **design confound** $\longrightarrow$ poor internal validity
- Experiments can implement _____ controls that prevent confounds.

Discuss: Previously, we discussed a design that gives a statistical control, in contrast to the design control we discuss now - what design was that, and in what context did that control work?

3.4.1 Variability

- _____ **Variability** is when a confound changes in a _____ a predictable manner *with* the independent variable
- **Unsystematic Variability** is when a confound changes in some way _____ or indifferent to the independent variable. It may occur in a seemingly random fashion.
  - The result is that there may be variability in the outcome, but if it happens _____ across all the conditions, it is not as much a concern.

Discuss: Based on our discussions in this lecture so far, which of these seems worse for us, and why?

3.4.2 Selection Effects

- **Selection Effects** occur when there is systematic variance of some factor across the conditions of the independent variable
  - This ends up being a confound $\rightarrow$ is the change in outcome due to this variance of a third variable or because of the condition? We _____ know
  - *Example:* In a treatment study on memory treatments, we assign some people to treatment A and some to treatment B. However, the group with treatment A comprises of mostly older individuals, whereas group B is mostly younger. Age may play a role in resulting memory impairment.
- This can be largely prevented by using **random assignment**, where members of the _____ are sort at random into conditions of the IV
- Using **matched groups** is another method to reduce the _____ of selection effects
  - This is where participants are measured on some third variable that may confound and ranked. Then each _____ is randomly placed in an alternating group
  - A major difficulty of this method is you must accurately identify what _____ should be matched on.

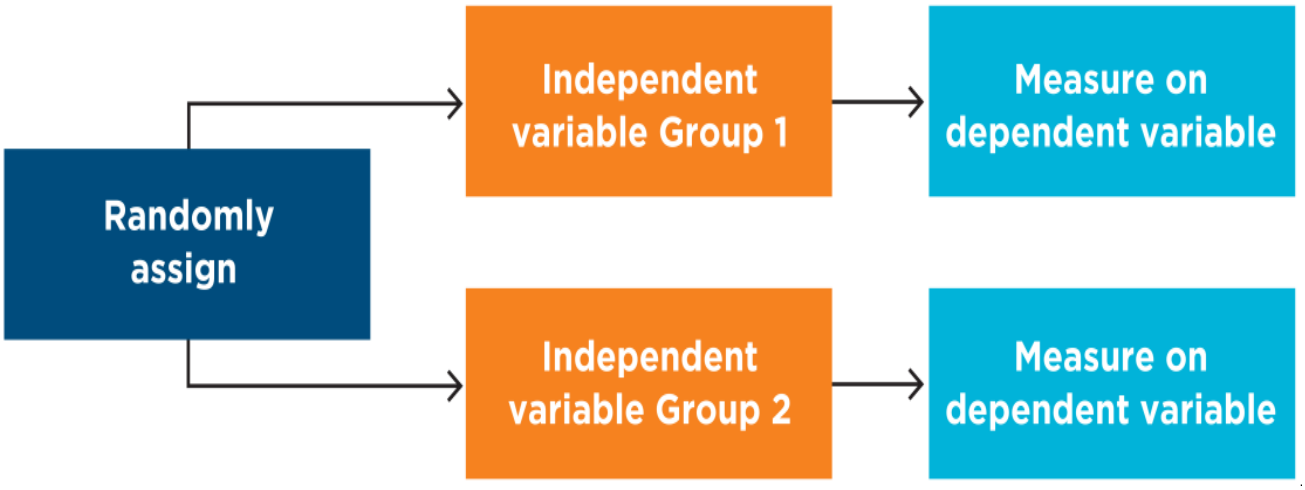
4 Independent-groups Design

4.1 Overview

- **Independent-groups designs** are those in which the participants of each condition are entirely _____ from one another
  - A.K.A. as between-subject or between-groups

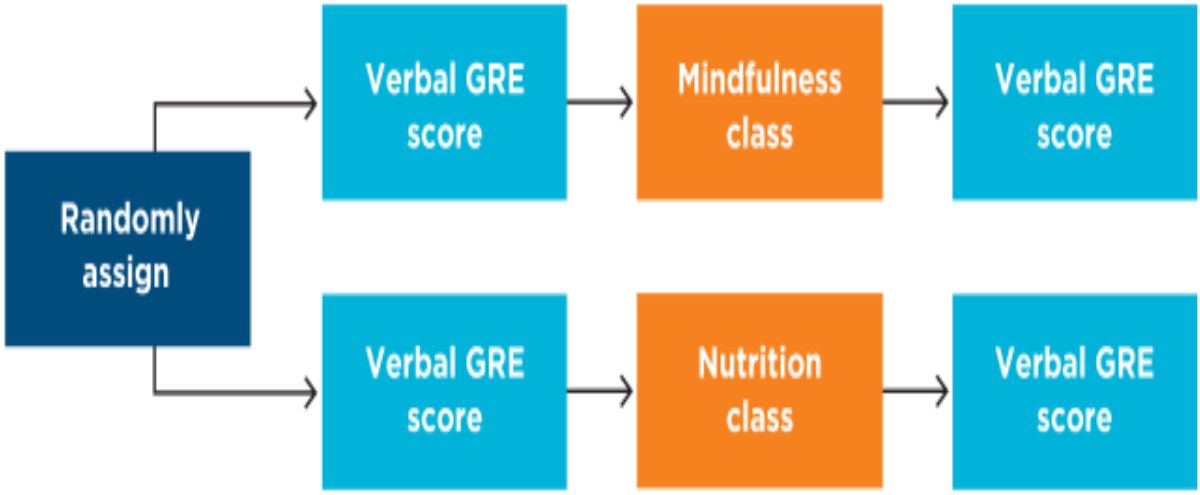
4.2 Posttest-only

- **Posttest-only design:** Separate groups of participants, assigned to different conditions, are measured only after the assignment and _____ has occurred.



4.3 Pretest/Posttest

- **Pretest/Posttest design:** Separate groups are tested on the outcome _____ the random assignment, but before *and* after the intervention



4.4 Comparison

- Both the posttest-only and the pretest/posttest designs are perfectly _____ experimental designs to investigate causal claims
- The primary difference is:
  - Posttest-only _____ that random assignment produces equivalent groups prior to interdention
  - Pretest/Posttest does not make this assumption and gives measures to validate
- Ideally, pretest/posttest is the "better" design as it provides more _____ and data - but we may use posttest-only for practicality

? I measure a sample of folks on their extraversion and happiness using two continuous measures and then measure them again in a year - what design is this?

- (●) A) Pretest/Posttest
- ( ) B) Bivariate correlational
- ( ) C) Posttest-only
- ( ) D) Multivariate correlational

Explanation:

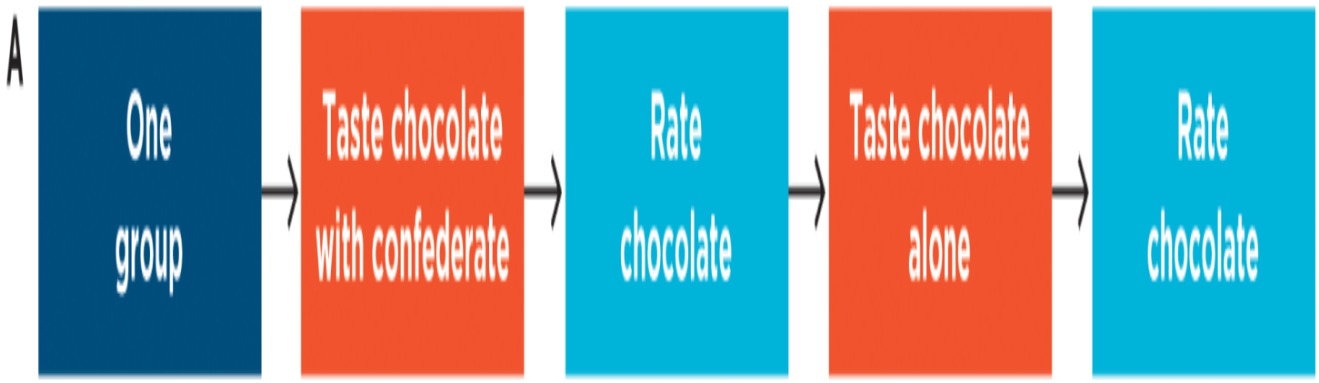
5 Within-groups Design

5.1 Overview

- **Within-groups designs** are when the _____ group of individuals is exposed to each condition
  - This is in contrast to the separate groups of the prior designs

5.2 Repeated Measures

- As the name would suggest, **repeated-measures** is when measurement on the outcome(s) of interest is taken after exposure to each condition, which happens _____



5.3 Concurrent Measures

- **Concurrent measures** is when both "conditions" are displayed to the participant at the _____ time and the measure of interest is some function of the behavior or affect towards the conditions
  - *Review:* Remember Harlow's _____ ?
- Depending on the conditions, this may not be a viable strategy

5.4 Causal Criteria in Within-Groups

- Do we have covariance?
  - Manipulate and measure two separate variables to see their _____

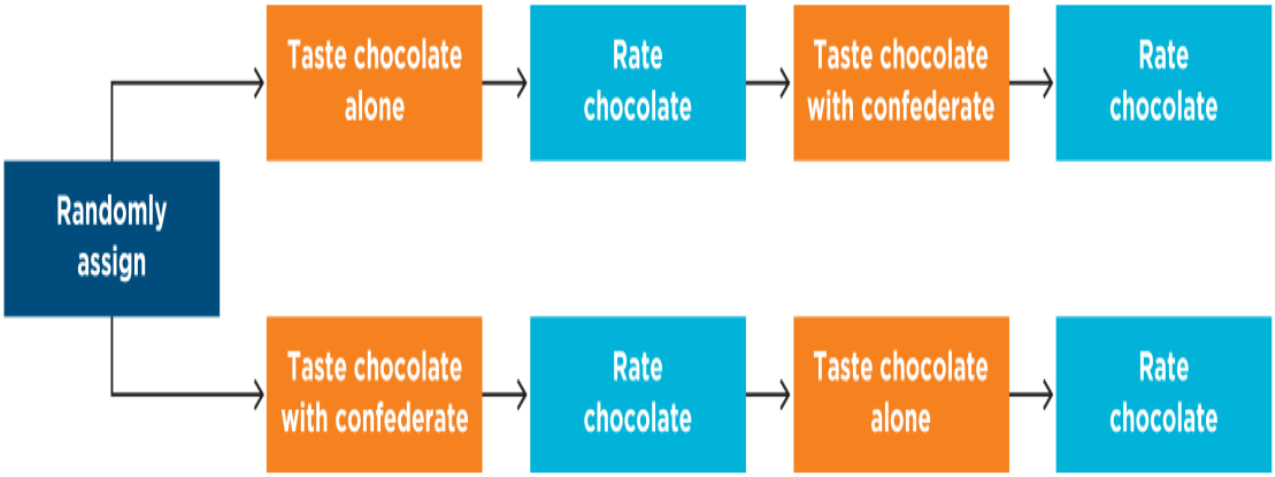
- Do we have temporal precedence?
  - Each measure comes _____ the intervention or manipulation

5.4.1 Internal Validity

- Participants _____ for themselves, i.e., random assignment isn't necessary
  - Because the individual is the same across conditions, we don't need to ensure they are _____ on some variable like with between groups
- However, we do need to be concerned with **order effects**, where exposure to one condition can have an _____ on the next reaction to a different condition
  - **Practice effects:** When someone becomes more _____ at a task due to practice on the prior conditions
  - **Fatigue effects:** When someone becomes _____ skilled at a task due to tiring from repeated testing
  - **Carryover effects:** When the effect of the previous condition is still _____ at the start of the next condition

📢 Discuss: Give an example of a situation where you think practice effects could occur on a certain measure

- Order effects are best prevented by **counterbalancing**, that is assigning some individuals to one _____ of conditions, and assigning the others to a separate order
  - **Full counterbalancing:** when _____ possible orders are done
  - _____ **counterbalancing:** when only some possible orders are represented



5.5 Advantages

- The "groups" are naturally equivalent, as the conditions are tested upon the same individual!
  - I.e., no concern for _____ effects, like in between-groups
  - We are able to express a strong _____ over all the various personal characteristics that could vary unsystematically
- Also requires _____ sample sizes for adequate power

5.6 Disadvantages

- As previously discussed, *counterbalancing* is necessary in order to _____ order effects from occurring
- We must attempt to "return" to baseline after each condition - but some interventions may have a _____ or prolonged effect on the participants, confounding the next condition
- We have to contend with the possibility that a participant _____ the hypothesis of a study, by virtue of seeing all the conditions, and changes their behavior in relation to this knowledge.
  - This is called a **demand characteristic**
  - *Review:* This is almost similar to "_____", as we discussed with self-report measures

? What validity is a demand characteristic a threat to?

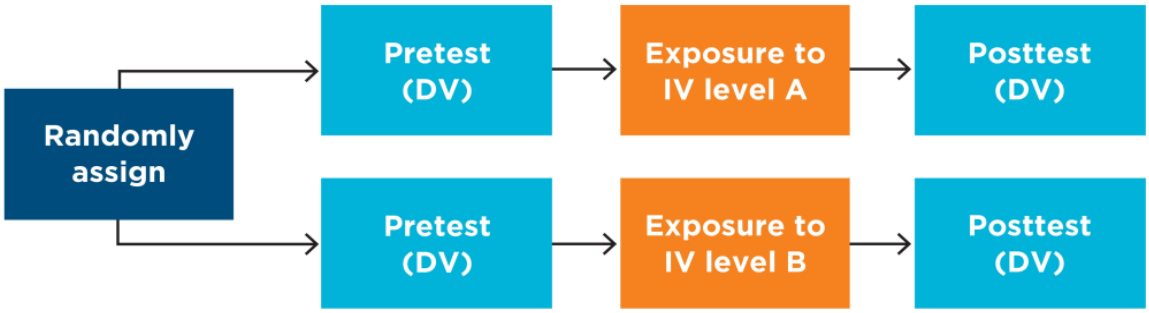
- (●) A) Construct validity
- ( ) B) Internal validity
- ( ) C) External validity
- ( ) D) Statistical validity

Explanation:

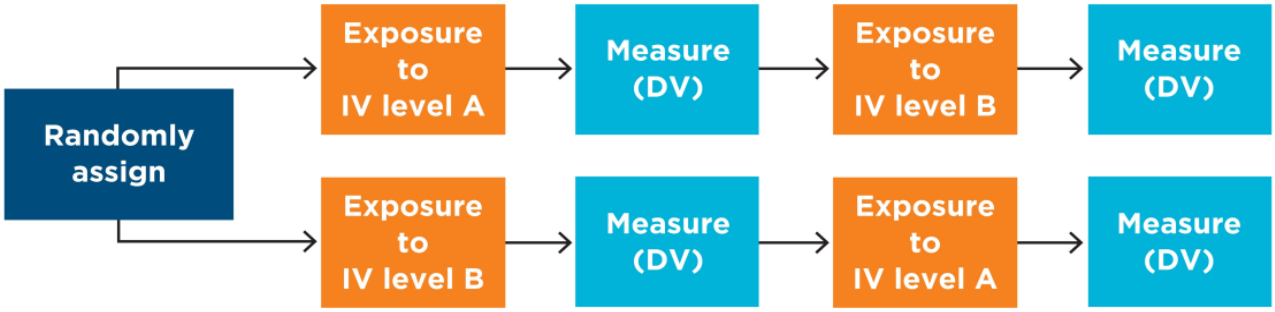
5.7 Pretest-posttest vs Repeated Measures

- The important difference is that whether conditions are _____ to different groups or the same people!

Pretest/posttest design



Repeated-measures design



6 Four Validities Critique

6.1 Construct Validity

- Dependent variables are likely to be _____ the same way we have previously discussed self-report and observational measures.

> Discuss: What were the 3 primary threats to observational design? Briefly describe them

- Independent variables are often based, at least initially, on whether they are _____ valid.
  - We may use a **manipulation check**, or an additional measure meant to ensure that the intervention had the _____ effect
  - This can be a good use for a **pilot study**
- All choices of variables to measure and manipulate should be guided by an understanding of the existing _____ in the area $\rightarrow$ we must assess whether the variables represent the constructs of the theory well

6.2 External Validity

- As previously discussed, sample _____ is the primary factor to consider in whether a sample is representative a specified population, and thus, whether results are generalizable.
- Extrapolating results to other, similar situations, requires _____ research
- Typically, experimental research tends to have worse external validity due to tight controls - the results aren't applicable to "normal" life - but they may be _____ with strong correlational designs (i.e., longitudinal or multiple regression) to help with this

Discuss: What about sample sizes? What role will they play in external validity or the other validities?

6.3 Statistical Validity

- Just like previous statistical validity discussion, we are concerned with 4 aspects:
  - Effect _____ : magnitude of effect
  - Precision: What confidence do we have that this effect falls in a narrow range of values
  - Replication: When a study is replicated, are statistics _____.
  - Significance: Were these results just a fluke of _____?

6.4 Internal Validity

- Covered in the previous sections, but this is the key question:
  - "Are the effects found here some _____ error, a third variable, or genuinely the intervention?"

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms